

Technical Proposal SERB

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Technical Details

Title: Context-Aware Privacy Preservation in the Internet of Things

1. Origin / Background of the Proposal:

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Context-awareness is the competency of a system or system component to accumulate information about its environment at any given time and adapt behaviors accordingly. Contextual or context-aware computing uses software and hardware to automatically gather and analyze data to drive responses.

Privacy protection remains a serious barrier to the large-scale deployment of context-aware and universal computing environments.

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An entity is a person, place, or thing that is considered significant to a user's engagement with an application, such as their location, time, actions, and preferences (Dey, 2001). The ability to use context information is known as context-awareness. A system is context-aware if it can extract, interpret and use context information and adapt its functionality to the current context of use (Byun & Chev5st, 2004). Those working in context-aware computing are familiar with the term. They believe that context is critical in their efforts to disseminate and transparently weave computer technology into our lives. Context-aware systems aim to acquire and use information about a device's context in order to provide services that are appropriate for the people, place, time, event, and so on.

To monitor and efficiently use resources, smart environments are fitted with various types of sensors and RFID. The IoT devices build a network by connecting via wireless connections and sharing data via edge networks[1]. The home automation system gathers data on power use and other user behaviors on a regular basis. Such information is extremely confidential and should not be shared. As a result, developing an effective privacy-preserving home automation system is critical. The user's identity, location privacy, and everyday activity are all protected by a privacy-preserving home automation system.

There is a growing demand for adaptable services based on context-aware apps. The efficiency of such services is largely determined by the user's context being acquired, organized, accessed, processed, and shared.

As a result, we present a set of strategies for developing a Privacy-Preserving IoT Model that addresses users' privacy concerns in context-aware ecosystems.

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2. Review of status of Research and Development in the subject

For Context-Aware Scalable Authentication, a probabilistic framework [1] is developed, which actively picks dynamic authentication techniques that satisfy security needs to be given passive factors. The prototype which is used for location classification needs location information after

every authentication. One of the disadvantages there is the possibility of forgetting the PIN of the user as it is used less.

Qingsheng Zhang et al. [2] provide a study on privacy of personal information on context awareness. This paper explains by using a specific computing context aware application which makes available suitable services for consumers. The privacy protection for context awareness is also discussed. The privacy role is explained which is capability of managing personal information of people in a supply way. Privacy role and drawing out of privacy revelation policies from consumer communication history is analyzed. A framework defined in [3] facilitates people to manage gathering personal information by different groups.

A strategy [4], [5] is discussed to control personal information for balancing the personal privacy concerns. The defined method is proved by a case study. When and how privacy issues come into the picture is discussed. Also it discusses how ubiquitous computing is better for privacy.

The experimental analysis of context aware mobile services are explained in [6]. This is accomplished using public key infrastructure, which allows web-enabled mobile devices to execute healthcare online services. Also the end to end security is defined. The paper gives a brief introduction to context aware web services where it defines context aware for web services and the limitation to security in web services. Then a literature survey is discussed. The performance analysis is done by encryption using the RSA algorithm. The output is considered with response time and hit ratio.

3.0 Work Plan:

Sl.No	Activity Planned	Approximate Time Taken (in weeks)
1	Survey and collection of user centric data	5
2	Application Development	5
3	Clustering of Data	2
4	Building a secure context aware privacy preserving mechanism using a) Anonymization b) Multi-Party Computation c) Homomorphic Encryption	9
5	Testing the developed Model	5
6	Validating the developed Model	6
7	Submission of the Project	4

	Total Time Taken (approx)	36 (3 Years)
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3.1 Methodology:

With their interconnected gadgets and services, pervasive computing environments offer seamless integration of digital infrastructure into our daily life. While current research is focused on how to connect new devices and develop useful applications to improve functionality, the security and privacy issues that arise in such environments have not been well investigated. Pervasive computing applications frequently utilize physical location and other context information about users and resources to enhance the user experience, whereas traditional distributed computing research aims to abstract away the physical location of users and resources.

The necessity to collaborate and share resources generates new sorts of user engagement as well as the interaction between the virtual and physical worlds. It's tough to tell the difference between physical and digital security in this situation.

Large amounts of data are collected from anywhere at any time in Internet of Things (IoT) systems, which may compromise people's privacy, especially when the systems are used in medical and everyday settings. Because most Internet technologies, communication protocols, and sensors are not built to enable IoT, the integration of smart devices into conventional Internet raises significant security concerns. Launching security/privacy attacks on IoT active systems, particularly Wearable Medical Sensor (WMS) devices, has been shown in recent research studies to result in catastrophic scenarios and life-threatening conditions. As a result, security vulnerabilities and privacy problems in the IoT should be thoroughly investigated.

As a result, the focus of this study is on a four-layer privacy authentication protocol for IoT end devices that do not have the capacity to properly identify the device of the request's sender, allowing some attacks to be mitigated. We looked at the protocol's security features for five common IoT risks and examined using the Blakley Sharing scheme to build a key generation and distribution system for safe communications between edge and end devices. The results of the experiments are expected to show that the proposed authentication protocol using the Blakley method is more efficient as the number of instructions increases in both fog structures and non-fog structures, indicating that the Blakley method is more flexible than the Schemes due to the increasing number of instructions indicating an increase in the number of nodes in the network.

Below are the steps towards building a secure context-aware privacy-preserving IoT Model.

1. Survey and Identification of user centric IoT systems for data collection.
2. Developing application to collect data.
3. Data Collection and clustering the data for privacy preservation.

(a) Experimentation and fixing the value of number of data clusters to value say K.

4. Building a secure context aware privacy preserving mechanism for each cluster using the following suitable techniques.

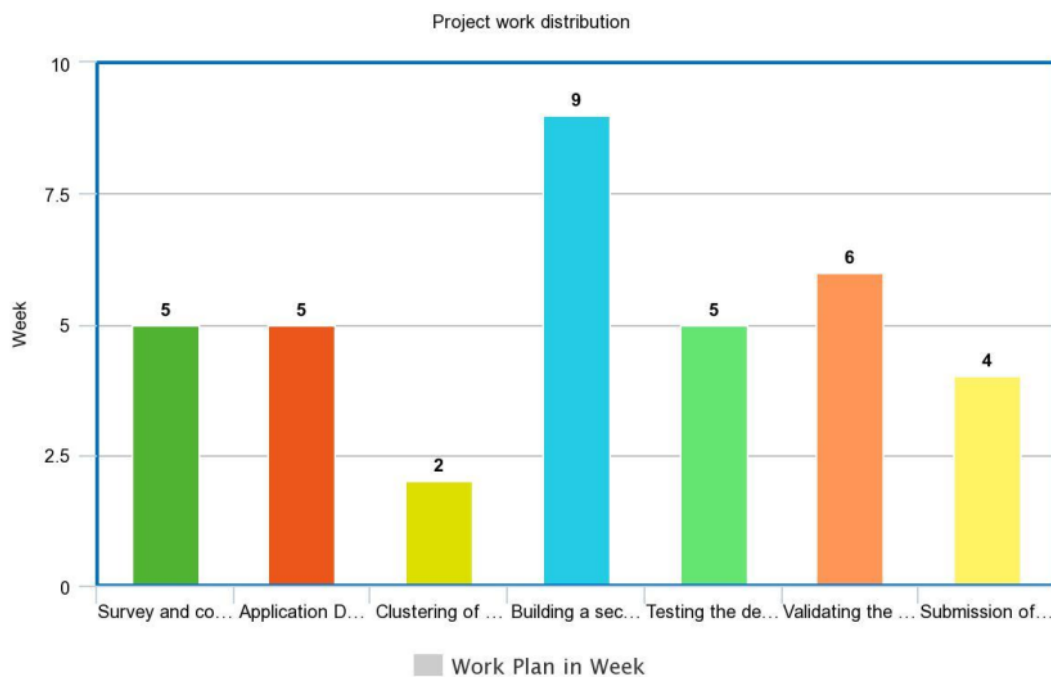
- (a) Anonymization
- (b) Multi-Party Computation
- (c) Homomorphic Encryption

5. Testing and validating / evaluating the proposed model.

The combination of above methods contribute towards development of Privacy-Preserving IoT Model

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3.2 Time Schedule of activities giving milestones through BAR diagram.



meta-chart.com

3.3 Suggested Plan of action for utilization of research outcome expected from the project.

In this paper, we propose a method for creating a trust index in the context of IoTs. Each item (human or machine) must have a diverse set of qualities and relationship. When describing the features of an object, attitudes should include information about the trust. Direct measurement or reasoning based on credible data from a variety of sources can be used to gain this information. The relationship should reflect the items' static and/or transitory relationships. Based on

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attributes and relationships, we may be able to establish what kinds of attributes and relationships are required to ensure trust in the application process.

To ensure that no equipment has been compromised that could compromise an individual's personal information or reputation, as well as that their health or even life is not risked if, for example, a medical device is instructed to give an overdose of medication to a patient. Although work is being done to guarantee that the various stakeholders in the Internet of Things ecosystem apply security measures appropriate to the dangers posed by these devices, if we are to establish a secure Internet of Things environment, reliable and effective protections must be built-in.

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3.4 Environmental impact assessment and risk analysis. (Maximum ½ page)

NIL

4. Expertise:

4.1 Expertise available with the investigators in executing the project:

Sl.No	Investigators Name	Number or Publication	Domain	Skill set
1	Dr.Thyagaraju G S	54	Context-Aware Computing, Machine Learning, IoT, Deep Learning, Artificial Intelligence, and Data Analytics.	Algorithm Design, Python Programming, App Development, Data Preparation, Data Analysis and IoT System Design and Development.
2	Prof.H Manoj T Gadiyar	13	Context-Aware Privacy, Cloud Computing, Artificial Intelligence, Augmented Reality, Internet of Things,	Algorithm Design, Python Programming, Knowledge on Object Orientation, Software Development Life Cycle Design.

4.2 Summary of roles/responsibilities for all Investigators:

Sl.No	Name of the Investigators	Roles / Responsibilities
1	Dr. Thyagaraju G S	Principle Investigator
2	H. Manoj T. Gadiyar	Co-Investigator

4.3 Key publications published by the Investigators pertaining to the theme of the proposal during the last 5 years

Published a paper titled, “Context Aware Privacy for Consumers using IoT and Cloud Computing” in PENSEE International Journal ISSN: 0031-4773 Volume 51, Issue 2 Page No: 1181-1193 indexed in Scopus list for the year 2021.

Published a paper titled “Context Sensitive Privacy Algorithm Method for Users in Cloud” in JNNCE Journal of Engineering & Management (JJEM) ISSN: 2582-0079.

Published paper titled “Privacy and Security issues in IoT based Smart Home Applications' ' in International Journal of Engineering Research & Technology: ICRTT 2018 Paper ID: IJERTCONV6IS15010.

Published paper titled “Conceptual Design of Driver Assistance Application using IOT” in Mantech Publications in Journal of Software Engineering and Software Testing , Volume 2 Issue2 ,Pae1-7 , 2017

Published paper titled “An Approach to the Detection of Retinoblastoma based on Apriori Algorithm” in International Journal on Recent and Innovation Trends in Computing and Communication, Volume: 5 Issue: 6,ISSN: 2321-8169, page 733 – 738 , June 2017.

Published paper titled “.A Smart Home Application for Resident Activity Prediction”, in Journal of Android and IOS Applications and Testing, Volume 2 Issue 2, May-August 2017,Page 1-8 © MAT Journals 2017

Published paper titled , Prediction Techniques in Internet of Things (IoT) Environment: A Comparative Study”, in International Journal on Recent and Innovation Trends in Computing and Communication, ISSN: 2321-8169,Volume: 5 Issue: 7 ,581 – 590, July 2017 (Refereed Journal)

Published paper titled “ IOT BASED SMART AND SAFE DRIVING APPLICATION”, in International Journal of Recent Innovation in Engineering and Research, Scientific Journal Impact Factor -3.605 by SJIF,e- ISSN: 2456 – 2084, Volume: 02 Issue: 02 February – 2017 (IJRIER, page 18 – 24)

Published paper titled “ACTIVE AND ASSISTED LIVING: A COMPREHENSIVE REVIEW OF ENABLING TECHNOLOGIES AND SCENARIOS”, Volume 9, No. 1, January-February 2018, **International Journal of Advanced Research in Computer Science** .

Published paper titled, “Privacy and Security Issues in IoT Based Smart Home Application” in International Conference on “Recent Trends in Technology” ICRTT -18th May 2018

An Intelligent Automated Door Surveillance System using Smart Camera” published in International Journal of Computer Application, International Journal of Computer Applications 181(42):13-15, February 2019. , Digital Library URI : <http://www.ijcaonline.org/archives/volume181/number42/30342-2019918478> ISBN : 973-93-80899-78-0 <https://www.ijcaonline.org/archives/volume181/number42/sharath-2019-ijca-918478.pdf>

Published paper titled , “Ambient Assisted Living: A Research on Human Activity Recognition and Vital Health Sign Monitoring using Deep Learning Approaches”, Retrieval Number: F11110486S419/19©BEIESP DOI: 10.35940/ijitee.F1111.0486S419 , Published By: Blue Eyes Intelligence Engineering & Sciences Publication , pp : 531 – 540. International Journal of Innovative Technology and Exploring Engineering (IJITEE) ISSN: 2278-3075, Volume-8, Issue-6S4, April 2019.

Published paper titled “An Automated Approach for Detecting Diabetic Retinopathy Using Bag of Words Method” , Journal of Seybold Report, VOLUME 15, ISSUE 8 August 2020, ISSN: 1533-9211, PP 545-550,

Published paper on “Chatbot for Car Museum” in Journal of Computer Science Engineering and Software Testing: 6 pp. 17-20 (1), Published on February 1, 2020.

Published paper on “ AI and Cloud based Smart Farm Assist in Kannada for Coconut Farmers” International Journal of Engineering Applied Sciences and Technology (IJEAST), 2020 Vol. 5, Issue 4, ISSN No. 2455-2143, Pages 372-37.

Published survey paper , “Context Aware Privacy for Consumer using IoT and Cloud Computing” in PENSEE International Journal ISSN: 0031-4773 Volume 51, Issue 2 Page No: 1181-1193 indexed in Scopus list for the year 2021.

Published a paper titled “DETECTING ANOMALOUS ACTIVITIES IN SPATIO-TEMPORAL REGION” in International Journal of Engineering Applied Sciences and Technology (IJEAST), Vol. 6, Issue 1, ISSN No. 2455-2143, Pages 441-444, Published Online during May 2021.

Published paper titled “License Plate Detection and Recognition using Convolution Neural Network” in Journal of Network Security Computer Networks, e-ISSN:2581-639X, Volume 7, Issue 1, Year 2021.

Published paper titled "AI and Cloud based Smart Farm Assist in Kannada for Coconut Farmers" in *International Journal of Engineering Applied Sciences and Technology (IJEAST)*, 2020 Vol. 5, Issue 4, ISSN No. 2455-2143, Pages 372-37.

4.4 Bibliography

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3. Debiao He, Neeraj Kumar, Jong-Hyouk Lee, Senior Member, IEEE "Secure Pseudonym-based Near Field Communication Protocol for the Consumer Internet of Things". IEEE Transactions on Consumer Electronics, Vol. 61, No. 1, February 2015.
4. Mark Vinkovits, Erion Elmasllari "Anonymous networking meets real-world business requirements" 2012 IEEE 11th International Conference on Trust, Security and Privacy in Computing and Communications.
5. Ruhma Tahir, Hasan Tahir, Klaus McDonald-Maier, *Anil Fernando "A Novel ICMetric Based Framework for Securing the Internet of Things" 2016 IEEE International Conference on Consumer Electronics (ICCE).
6. Stelios Sotiriadis, Euripides G.M. Petrakis, Stefan Covaci, Paolo Zampognaro, Eleni Georga, Christoph Thuemmler "An architecture for designing Future Internet (FI) applications in sensitive domains: Expressing the Software to data paradigm by utilizing hybrid cloud technology" 978-1-4799-3163-7/13. 2013 IEEE.
7. Virgilio A.F. Almeida Federal University of Minas Gerais, Brazil Danilo Doneda • Rio de Janeiro State University Marília Monteiro Public Law Institute of Brasília "Governance Challenges for the Internet of Things". Published by the IEEE Computer Society 1089-7801/15/\$31.00 © 2015 IEEE.
8. Yang Jin-cui, PANG hao, ZHANG Xin "Enhanced mutual authentication model of IoT". December 2013, 20(Suppl. 2): 69–74
9. Young Seog Yoon/ ETRI Hyun-Woo Lee / ETRI "What kinds of "Things" Should be Considered in Designing Trust for IoTs Applications?: a Multidisciplinary Review". 978-1-4673-7116-2/15. 2015. IEEE.
10. Campbell, R., Al-Muhtadi, J., Naldurg, P., Sampemane, G., Dennis Mickunas, M. (2003). Towards Security and Privacy for Pervasive Computing. In: Okada, M., Pierce, B.C.,

Scedrov, A., Tokuda, H., Yonezawa, A. (eds) Software Security — Theories and Systems. ISSS 2002. Lecture Notes in Computer Science, vol 2609. Springer, Berlin, Heidelberg. https://doi.org/10.1007/3-540-36532-X_1

11. (2019). A Group Authentication Protocol on Multilayer Structure for Privacy-Preserving IoT Environment. *Engineering and Technology Journal*, 37(5A), 172-180. doi: 10.30684/etj.37.5A.4

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5. List of Projects submitted/implemented by the Investigators

(All the Investigators should list out details of the Projects submitted, implementing and completed by them. The list should start with the Projects implemented by the Principal Investigator, followed by Co-PI1, Co-PI 2 etc.)

5.1 Details of Projects submitted to various funding agencies:

Sl.NO	Title	Cost in Lakh	Month of Submission	Role as PI/Co-PI	Agency	Status
Not Applicable						

5.2 Details of Projects under implementation:

Sl.No	Title	Cost in Lakh	Start Date	End Date	Role as PI/Co-PI	Agency
Not Applicable						

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6. List of facilities being extended by parent institution(s) for the project implementation.

6.1 Infrastructural Facilities

Sl.No	Infrastructural Facility	Yes/No/ Not required Full or sharing basis
1	Workshop Facility	Yes
2	Water & Electricity	Yes

3	Laboratory Space/ Furniture	Yes
4	Power Generator	Yes
5	AC Room or AC	Yes
6	Telecommunication including e-mail & fax	Yes
7	Transportation	Yes
8	Administrative/ Secretarial support	Yes
9	Information facilities like Internet/Library	Yes
10	Computational facilities	Yes
11	Animal/Glass House	No
12	Any other special facility being provided	NA

6.2 Equipment available with the Institute/ Group/ Department/Other Institutes for the project:

Equipment Available with	Generic Name of Equipment	Model , Make and Year of Purchase	Remarks including accessories available and current usage of equipment
PI and his group	NA		
PI's Department	NA		
Other Institute(s) in the region	NA		

7. Name and address of experts/ institution interested in the subject / outcome of the project

Name and address of the Experts:

1. Dr. R. H Goudar, Associate Professor, Department of CSE, Visvesvaraya Technological University, Belagavi, Ph: 8971584906, Email: rhgoudar.vtu@gmail.com
2. Dr. Santhosh L Deshpande, Professor & Head, Department of CSE, Visvesvaraya Technological University, Belagavi, Ph: 9448428785, Email: rldeshpande@vtu.ac.in
3. Mr. Ramprasad, CTO & Director, VRAIO Software Solutions Pvt Ltd, Bangalore Ph: +91-97387 35077, Email: rama@vraio.in

Expected Project Outcome:

1. Establishment of Context Aware Communication between IoT Devices and Cloud.
2. Providing more secured communication between IoT devices and Cloud using the proposed Multilayer Context aware Authentication Protocol.
3. Improved Performance in terms of security and privacy.

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Approaches", International Journal of Innovative Technology and Exploring Engineering, 2019

Publication

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Dong Hu, Hansheng Lao. "Privacy Research on Ubicomp Computing with Neural Cryptography", 2008 The 3rd International Conference on Grid and Pervasive Computing - Workshops, 2008

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Young Seog Yoon, Hyun-Woo Lee. "What kinds of "things" should be considered in designing trust for IoTs applications?: A multi-disciplinary review", 2015 International Conference on Information and Communication Technology Convergence (ICTC), 2015

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